

Solicitation Number: USSOCOM RFI Technical Experimentation Event 25-3: Human Machine Teaming

Notice Type: Special Notice

TYPE: A–Research and Development

NAICS: 541715 Research and Development in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology)

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Response Date: 26 May 2025, 12:00 Noon Daylight Savings Time (DST)

Synopsis:

A. INTRODUCTION: Technical Experimentation (TE)

This Request for Information (RFI) is NOT a solicitation for proposals, proposal abstracts, or quotations. The purpose of this RFI is to solicit technology experimentation candidates from Research and Development (R&D) organizations, private industry, and academia for inclusion in future experimentation events coordinated by the U. S. Special Operations Command (USSOCOM). USSOCOM invites industry, academia, individuals, and Government labs to submit technology experimentation nominations addressing innovative technologies identified in paragraph 4 below. The intent is to provide participants with the opportunity to gain Special Operations Forces (SOF) insight/perspective on participant technologies.

Technical experimentation will explore emerging technologies, technical applications, and their potential to provide solutions for future SOF capabilities.

This RFI is for TE 25-3. This event will occur over the period of one week as shown below.

Dates/Theme/Location:

8 September - 12 September 2025: Human Machine Teaming – at the Emerging Technology Institute (ETI), Red Springs, North Carolina (NC).

B. OBJECTIVE:

1. Technology experimentation events provide an opportunity for technology developers to interact with operational personnel to determine how their technology development efforts and ideas may support or enhance SOF capability needs. The environment facilitates a collaborative relationship between Government, academia, and industry to promote the identification and assessment of emerging technologies.

2. The deadline for nomination package(s) is **26 May 2025 at 12:00 Noon DST**. After review of the TE nomination submissions, the Government may invite select candidates to demonstrate their technologies at the USSOCOM sponsored TE event. Experiments will be conducted from 8-12 September 2025, in Red Springs, NC and will explore emerging technology solutions and revolutionary improvements in relevant technologies. Materiel solutions brought to the event should be between a Technology Readiness Level (TRL) of 3 and 6 or higher as stated in paragraph 4. Proposed experiments may be between a half day and two days in duration and may be

conducted in unimproved expeditionary-like conditions. There is no intention on the part of USSOCOM to purchase or procure equipment based solely on participation in the TE.

3. Experimentation Focus: The primary intent of this event is to highlight technologies that support Human Machine Teaming.

4. Technology areas to explore during the event include the following:

4.1 Modular, National Defense Authorization Act (NDAA) compliant, low Size, Weight, and Power-Cost (SWaP-C) UxS Solutions.

4.1.1 Modular components.

4.1.2 Low cost UxS

4.1.3 Attritable / Disposable UxS.

4.1.4 Rapidly produced UxS systems.

4.1.5 Agnostic to analytic or control software.

4.1.6 Kinetic or non-kinetic modular payloads.

4.1.7 UxS production or manufacturing at the edge.

4.1.8 Group 1 or 2 UAS-mounted weaponry.

4.1.9 Low SWaP jam resistant GPS antennas.

4.2 Multi-Modal, Multi-Domain Unmanned Systems and capabilities

4.2.1 Multi-Modal UxS e.g., single system that is air, land, and sea capable.

4.2.2 Low cost, non-attributable multi-domain sensors and payloads with non-persistent (volatile) firmware and or remote zeroizing function e.g., Signals intelligence (SIGINT), Electronic Warfare (EW), Imagery Intelligence (IMINT), Geospatial Intelligence (GEOINT), Electronic Intelligence (ELINT), Measurement and Signature Intelligence (MASINT).

4.2.3 RF deception and decoy payloads, e.g. SIGINT, ELINT, and MASINT.

4.2.4 Low SWaP, novel, wideband, multi-element, phased antenna arrays.

4.2.5 Operations without conventional Radio Frequency (RF) communications.

4.2.6 Biomimicry, nature inspired designs.

4.2.7 Automatic Target Recognition (ATR) algorithms agnostic to custom data sets.

4.2.8 Novel ATR training capabilities, no-code/low-code solutions for training at the edge by operators.

4.3 Edge processing capacity

- 4.3.1 Innovative body worn hardware with low SWaP and acceptable operating temperatures.
- 4.3.2 Field Programmable Gate Arrays (FPGAs)
- 4.3.3 Quantum computers
- 4.3.4 Fanless computers
- 4.3.5 Quantum storage and memory
- 4.3.6 Qubit Processors
- 4.3.7 Innovative manufacturing of computer processing chips that support edge processing.
- 4.3.8 Edge Intelligence
- 4.3.9 Internet of Everything (IoE)
- 4.3.10 Edge security
- 4.3.11 5G and Beyond Edge Computing
- 4.3.12 Distributed computing capabilities
- 4.4 Heterogeneous teaming and UAS Swarming capabilities.
 - 4.4.1 UxS common controller
 - 4.4.2 UxS common control architectures.
 - 4.4.3 AI-enabled UAS swarming algorithms
 - 4.4.4 Pairing manned and unmanned systems with collaborative alerts on articles of interest in the operational environment for increased situational awareness.
 - 4.4.4.1 Machine Learning and cooperation/collaboration
 - 4.4.4.2 Optimized air/vehicle/vessel engagement
 - 4.4.4.3 Cooperative re-tasking and decision support
 - 4.4.4.4 Launch, recover, cooperate with UxS.
 - 4.4.4.5 Unmanned system networking.
- 4.5 Autonomous operations for GPS-denied environments
 - 4.5.1 Alternative navigation for small UxS.
 - 4.5.1.1 Noncooperatively use RF signals of opportunity from space based or terrestrial emitters for alternative position fixing.
 - 4.5.1.2 Visual-Based Navigation (VBN)
 - 4.5.1.3 Vision-Aided Navigation (VAN)
 - 4.5.1.4 Terrain Referenced Navigation (TRN)
 - 4.5.1.5 Visual Inertial Odometry (VIO)
 - 4.5.1.6 Simultaneous Localization and Mapping (SLAM)

- 4.5.1.7 Magnetometry
- 4.5.1.8 Light Detection and Ranging (LiDAR) from US and non-covered foreign country origins only.
- 4.5.1.9 Radio Detection and Ranging (RADAR) Navigation
- 4.5.1.10 Inertial Navigation System / Inertial Measurement Unit
- 4.5.1.11 Celestial Navigation
- 4.5.1.12 Photogrammetry-derived navigation at the edge
- 4.5.1.13 Alternate Position Navigation Timing (PNT) (Using timing for navigation).
- 4.5.2 Novel solutions to allow small unmanned systems to navigate in GPS denied environments in real time without reach back to secondary control units.
- 4.6 First Person View (FPV) UxS and components from non-covered foreign countries or US manufacturers only.
 - 4.6.1 FPV Goggles for Military Applications
 - 4.6.2 Modular FPV UAS
 - 4.6.3 FPV targeting solutions
 - 4.6.4 FPV Tak Plug-Ins
 - 4.6.5 Encrypted C2 links
 - 4.6.6 US-made video transmitters (VTX)
- 4.7 Data Transport
 - 4.7.1 A single multi-link, multi-band receiver antennae capability, with multiple integrated receivers for all direct data feeds from the space, aerial, and terrestrial sensors.
 - 4.7.2 Cross-edge node/mesh-like data federation capability that synchronizes data & products from one edge node to another when cloud connectivity is not practical.
 - 4.7.3 Multi-band communication relays that use the widest possible part of the electro-magnetic spectrum.
 - 4.7.4 Acoustic communications for UxS operations
 - 4.7.5 Visible/infrared communications for UAS operations
 - 4.7.6 Fiber optic communications for UxS operations.
 - 4.7.7 Laser photonic communications (The use of lasers for data transmission).

4.8 Voice and Data Processing

- 4.8.1 Artificial Intelligence (AI) for digital information processing.
- 4.8.2 Data integration and interoperability across platforms.
- 4.8.3 Server hardware in a scalable edge compute form factor (laptop, lightweight transit case server, vehicle rack mounted) capability to host cloud services on local hardware supporting solutions for theater strategic to tactical units in light, medium, and heavy SOF formations for both garrison and austere conditions.
- 4.8.4 Server hardware with persistent store (data fabric) capability that can support data and products within the SCI, Collateral, Collateral Coalition, and CUI security classification domains.
- 4.8.5 Data compression capabilities
- 4.8.6 Provide near-instantaneous voice and data communications.
- 4.8.7 Enable crossband waveform integration to link disparate networks, including support for proprietary waveforms within VHF, UHF, L-Band, S-Band, and C-Band.
- 4.8.8 Capable of operating on multi-level security networks.
- 4.8.9 Support Combined Joint All Domain Command and Control (CJADC2) by providing a shareable common operating picture and data fabric with data bridges for bi-directional data flow.

4.9 UxS Human-Machine Interface capabilities

- 4.9.1 AI algorithms to enhance UxS autonomy at the platform level.
- 4.9.2 Novel approaches to RF-based control architectures
- 4.9.3 Non-RF control architectures
- 4.9.4 Novel, multi-modal control methodologies for single and multi-agent UxS control.
- 4.9.5 Novel, multi-modal control methodologies for End User Device (EUD) based UxS controllers.
- 4.9.6 Bi-directional human machine state sharing.
- 4.9.7 Novel approaches to UxS user interfaces.
- 4.9.8 Novel augmented reality displays.
- 4.9.9 Any visual devices that integrate technology and machine learning to enhance the understanding of the user.
- 4.9.10 Hands-free command interface, i.e., voice control of UxS.
- 4.9.11 AI enabled Common Operating Picture (COP)
- 4.9.12 Multi-Sensor exploitation and integration, sensor fusion.
- 4.9.13 Automatic Threat Recognition (ATR) integration

- 4.9.14 Internet of Things (IoT) sensor integration
- 4.10 Unmanned Aerial Systems (UAS) capabilities
 - 4.10.1 Small Unmanned Air Systems (SUAS) charging hives.
 - 4.10.2 Solar/wind/hydro/combustion powered recharging capability.
 - 4.10.3 Parasitic charging capabilities.
 - 4.10.4 In-flight recharging, i.e., inductive charging.
 - 4.10.5 Novel propulsion capabilities.
 - 4.10.6 Security features for concealment/defense.
 - 4.10.7 SUAS battery life extensions.
 - 4.10.8 SUAS durable and efficient solar power options.
 - 4.10.9 SUAS employment (Infil/placement/exfil) of payload and payload data.
 - 4.10.10 Carrying, deployment, retrieval mechanisms.
 - 4.10.11 Automated intercept/retrieval of payload data.
 - 4.10.12 Tethered SUAS.
- 4.11 Unmanned Ground Vehicles (UGV)
 - 4.11.1 Robust person packable and portable robotic systems.
 - 4.11.2 Multipurpose legged, wheeled, or tracked robotic systems.
 - 4.11.3 Uniquely designed robotics for subterranean environments or complex navigation.
- 4.12 Counter UxS capabilities.
 - 4.12.1 Operator portable UxS detection, localization, and classification capabilities that operate in dismounted transit scenarios with minimal or no electromagnetic emission.
 - 4.12.2 Novel operator portable UxS defeat capabilities.
 - 4.12.3 Vehicle mounted low SWaP UxS detection and defeat capabilities.
- 4.13 Counter Ubiquitous Technologic Surveillance (C-UTS) Technologies
 - 4.13.1 Software or hardware capable of denying or deceiving UTS technologies.
- 4.14 Blue force awareness technology
 - 4.14.1 Devices or technology to distinguish friend from foe on UxS platforms.

- 4.14.2 Devices or technology to distinguish friendly and partner forces visually or digitally in a GPS / Communications contested environment.
- 4.15 High Altitude Sensor Payloads
 - 4.15.1 Lightweight man-portable systems.
 - 4.15.2 Chambered tested to stratospheric conditions from 60,000-80,000' Mean Sea Level (MSL).
 - 4.15.3 Low power consumption, ability to turn on or off.
 - 4.15.4 Payloads <6 lbs. for station seeking applications above 60,000 MSL.
 - 4.15.5 Payloads <16 lbs. for non-station seeking applications above 55,000' MSL.
 - 4.15.6 Payloads <32 lbs. for non-station seeking applications above 45,000' MSL.
 - 4.15.7 Line of Site ELINT, COMINT, or Electronic Surveillance (ES) capability.
 - 4.15.8 Operate under Title 10 authorities.
 - 4.15.9 Direction finding / geolocation on RF data.
 - 4.15.10 Electro-Optical / Infrared Red with geolocation metadata and imagery no less than 2 pixels per meter at altitude.
 - 4.15.11 Volatile memory; no SD card.
 - 4.15.12 Can integrate with / integrated MPU5, SILVUS, or Starlink communications.
 - 4.15.13 Method of managing signal separation and co-channel interference.
- 4.16 High Altitude platforms
 - 4.16.1 Man portable systems
 - 4.16.2 Able to be employed by a two-person team in less than 15 minutes.
 - 4.16.3 Platform supports payload power needs.
 - 4.16.4 Payloads <12 lbs. for station seeking applications.
 - 4.16.5 Payloads <40 lbs. for non-station seeking applications.
 - 4.16.6 Capable of delivering High Altitude Launched Effects (HALES). Supports extended long range, secure Beyond Line of Sight (BLOS) communications for mobile land, maritime, and partner forces using hand-held radios.
 - 4.16.7 Support persistent ISR for periods greater than 24 hours.
- 4.17 Light-weight Air Launched Glide Munition
 - 4.17.1 Lightweight, precision, unpowered, air-launched glide munition to fill the gap for glide munitions smaller than the GBU-69 small glide munition.
 - 4.17.2 Portable as internal or external stores on any aerial system.
 - 4.17.3 Capable of reaching targets without an audible signature from standoff ranges of 100km.

- 4.17.4 In addition to U.S. military and threat GPS systems, the glide munition will use multiple type of the following GPS alternative systems for Navigation:
 - 4.17.4.1 Noncooperatively use RF signals of opportunity from space based or terrestrial emitters for alternative position fixing.
 - 4.17.4.2 Visual-Based Navigation (VBN).
 - 4.17.4.3 Vision-Aided Navigation (VAN).
 - 4.17.4.4 Terrain Reference Navigation (TRN).
 - 4.17.4.5 Visual Inertial Odometry (VIO).
 - 4.17.4.6 Simultaneous Localization and Mapping (SLAM).
 - 4.17.4.7 Magnetometry.
 - 4.17.4.8 Light Detection and Ranging (LiDAR) from US and non-covered foreign country origins only.
 - 4.17.4.9 Radio Detection and Ranging (RADAR) navigation.
 - 4.17.4.10 Inertial Navigation System / Inertial Measurement Unit.
 - 4.17.4.11 Celestial Navigation.
 - 4.17.4.12 Photogrammetry-derived navigation at the edge.
 - 4.17.4.12 Alternate PNT (using Timing for navigation).
- 4.17.5 The glide munition has an option for a semi-active laser capable of receiving in-flight targeting updates via two-way data link.
- 4.17.6 Warhead size will not exceed 10 pounds. Overall glide munition weight will not exceed 30 pounds.
- 4.17.7 Warhead types are interchangeable by the operator and should include:
 - 4.17.7.1 Airburst with adjustable height above ground setting
 - 4.17.7.2 Point detonation
 - 4.17.7.3 Penetration
 - 4.17.7.4 Armor piercing
- 4.17.8 Capable of operating in a -45 degrees Fahrenheit environment.
- 4.17.9 Utilize an explosive adhesive coloring over the fuselage and flight surfaces to enhance the explosion effects.
- 4.17.10 The munition will have a built-in data link to send its location and receive in-flight instructions to change targets or self-destruct.
- 4.18 Decoy Payload for UAS or high-altitude platforms
 - 4.18.1 The payload must have a reprogrammable RF signature.
 - 4.18.2 The payload must reflect RADAR.
 - 4.18.3 The payload must be expendable.
 - 4.18.4 The payload controller must be able to remotely zeroize the nonpersistent firmware.

5. Security/Classification Requirements: Respondents shall not submit classified information in the technology experimentation nominations.

6. **Safety Requirements:** All respondents shall review the TE Safety Guide (RFI Notice Attachment 3). Those respondents who are invited to demonstrate their technologies must complete a Deliberate Risk Assessment Worksheet (Department of Defense Form 2977) (RFI Notice Attachment 4) in accordance with MIL-STD-882E and the Department of the Army Techniques Publication No. 5-19 (ATP 5-19). Risk assessments shall be emailed directly to the **tech_exp@socom.mil** by **11 Aug 2025**. Respondents should include instructions that describe the safe operation of the device nominated for the experiment.

7. **Frequency Requirements:** **If your experiment will be radiating on a given frequency or frequency band, you MUST have prior approval to transmit on that frequency.** Prior approval includes compliance with Federal Communications Commission (FCC) Title 47, Part 15, or a **Special Temporary Authority (STA) from the FCC.**

a. You must have National Telecommunications and Information Administration (NTIA) frequency approval if your experiment includes Government-owned equipment, and you will be operating within a Federal Band.

b. **Respondents are advised to not wait for confirmation of selection/invitation to the event before requesting a STA from the FCC.**

c. Your authority to radiate **must be** emailed directly to **tech_exp@socom.mil** no later than **7 July 2025**.

All frequency questions shall be directed to the USSOCOM Technical Experimentation team at the email above. **Respondents shall NOT contact the ETI facility directly.**

8. **Human Subjects.** It is not anticipated that activities being conducted in this TE event will require the use of research or experimentation involving human subjects. Technology experiment submissions will be reviewed for potential research or experimentation involving human subjects. Any submission that is determined to potentially include research or experimentation involving human subjects will be required to adhere to DoD Instruction 3216.02 "Protection of Human Subjects and Adherence to Ethical Standards in DoD Supported Research" and ensure appropriate Institutional Review Board and DoD Human Research Protections Office approvals prior to conducting those activities.

9. **Other Special Requirements:** **DO NOT SUBMIT PROPOSALS OR MARKETING DEMONSTRATIONS. SUBMIT TECHNOLOGY EXPERIMENTATION NOMINATIONS ONLY. EXPERIMENTATION NOMINATION SUBMITTALS FOR THIS RFI WILL ONLY BE ACCEPTED UNTIL THE CLOSING DATE OF 26 May 2025, 12:00 Noon DST,** for the requirements stated above. No contracts will be awarded based solely on this announcement or any subsequent supplemental RFI announcements.

C. SUBMISSION INSTRUCTIONS:

Technology experimentation nominations shall be submitted electronically by creating a Scout Card in the Vulcan system at: vulcan-sof.com by the respective deadline. **Scout Cards should be created within the call for TE 25-3 Technical Experimentation:**

Human Machine Teaming. The Vulcan system is best accessed by using Firefox or Google Chrome browsers. Associated technology experiments with distinctly different uses or applications should have a separate nomination submitted by each respondent. USSOCOM personnel will review submissions to determine whether an experiment submission will be accepted for invitation to attend the TE event.

A complete submission consists of:

- Creating a Scout Card in the Vulcan system as described above.
- Completely filling in the Supplemental Sheet Version MAY2025 (RFI Notice Attachment 2) and attaching it to the Scout Card. **Note: The Supplemental Sheet has 31 sections that must be COMPLETED in its entirety. Incomplete supplemental sheets will be returned. Respondents shall not submit previous versions of the TE Supplemental sheet nor sheets from another DoD event.**
- Devices with radio frequency emissions must state the intended frequency or frequencies used by the device(s).
- An FCC STA, Experimental License, or NTIA document (for developmental radio frequency emitting devices). If neither is available at the time of submission, provide status of your FCC/NTIA request. **Note:** Those developers selected to participate and not transmitting in the ISM band must provide a current FCC STA, Experimental License, or NTIA document prior to operating their equipment. If you cannot provide a valid FCC ID or the afore mentioned temporary permission to transmit by the date of this event, you will not be permitted to transmit.
- Instructions on how to safely use the technology (as needed).
- If applicable, a picture of the device with a short description of the size (shows the dimensions or places the device next to a ruler, currency, or man-sized object for comparison).
- A signed Photo Release (RFI Attachment 5).

Selected respondents will be invited to participate in USSOCOM experiments. USSOCOM shall provide venues, supporting infrastructure, and Joint-SOF assessment personnel (operational and technical, based on availability of resources and written request as discussed above) at no cost to invited respondent(s). **All respondents' submission costs, travel costs, technology experiments, and experimentation associated costs will be at the respondents' expense.** The TE venue will only provide basic access to training areas or ranges (if approved and applicable) to conduct experiments, a facility to connect to the internet, basic venue infrastructure including frequency coordination/deconfliction, and shore power. Invited respondents must be prepared to be self-sufficient during the execution of their experiments and not dependent on venue resources.

Time and space will be made available for technology developers to conduct real-time modifications and updates to technologies. **Technology developers are advised to bring all tools and equipment necessary to present/operate their technology throughout the event.**

Additionally, select technologies of interest may be invited to participate in a focused assessment by the Defense Threat Reduction Agency (DTRA) Sensitive Assessment Division (TSX) as part of or immediately following the main TE event. Technologies selected for DTRA evaluation will be notified by the TE staff and further coordination will be communicated prior to the event.

D. BASIS FOR SELECTION TO PARTICIPATE:

Selection of respondents to participate shall be based on the extent to which the technology represents a potential capability increase to SOF.

Other considerations include:

- Technical maturity
- Relevance of or adaptability to military operations/missions
- Relevance to current operational needs
- Relevance to Event Focus Area

E. ADDITIONAL INFORMATION: All efforts shall be made to protect proprietary information that is clearly marked in writing. Lessons learned by USSOCOM from these experiments may be broadly disseminated, but only within the Government. If selected for participation in Technical Experimentation, developers may be requested to provide additional information that will be used in preparation for the experiments.

Future TE events with TENTATIVE themes/focus areas, locations, ANTICIPATED dates and RFI open periods are as follows:

Number	Theme/Focus Area	Location	Event Dates	RFI Open Period
TE 26-1	In development.	TBD	TBD	TBD

Technology developers invited to this event are encouraged to contact each other using the social media through USSOCOM Technical Experimentation. While USSOCOM Technical Experimentation has access to several social media, the preferred collaboration social media link is LinkedIn www.tinyurl.com/LinkedIn-SOCOMTE.

F. USE OF INFORMATION: The purpose of this notice is to gain information leading to Government/Industry collaboration for development of USSOCOM technology capabilities and to assist in accelerating the delivery of these capabilities to the SOF warrior. All proprietary information contained in the submission and technology experimentation shall be appropriately marked. The Government will not use proprietary information submitted from any one firm to establish future capability and requirements. During the event USSOCOM may broadcast technology experiments via video-teleconference back to USSOCOM and/or their operational components. Respondents to this RFI shall review and submit the photography approval form at Attachment 5.

G. SPECIAL NOTICE:

1. Federally Funded Research and Development Centers (FFRDCs) or contractor consultant/advisors to the Government will review and provide support during evaluation of submittals. When appropriate, non-Government advisors may be used to objectively review a particular functional area and provide comments and recommendations to the Government. All advisors shall comply with procurement integrity laws and shall sign non-disclosure statements. The Government shall take into consideration requirements for avoiding conflicts of interest and ensure advisors comply with safeguarding proprietary data. Submission in response to this RFI constitutes approval to release the submittal to approved Government support contractors.

2. There will be foreign attendees (military, government, or industry partners), who are interested in the capabilities being demonstrated at the TE event. Respondents are ultimately responsible for complying with all International Trafficking in Arms (ITAR) /Export Administration Regulations (EAR) requirements associated with their equipment. Before declaring their ITAR/EAR status, respondents to this RFI should verify the ITAR/EAR status of their technology by visiting the US Department of State Directorate of Defense Trade Controls (ITAR) website at: (https://www.pmddtc.state.gov/ddtc_public) and the US Department of Commerce Bureau of Industry and Security website (<https://www.bis.doc.gov/>) for more information on Export Administration Regulations (EAR). USSOCOM event organizers will restrict access as necessary to assist in protecting ITAR/EAR related technology demonstrations.

H. Per Federal Acquisition Regulation (FAR) 52.215-3, Request for Information or Solicitation for Planning Purposes (Oct 1997):

1. The Government does not intend to award a contract based on this RFI notice or to otherwise pay for the information.

2. Although "proposal" and "respondent" are used in this RFI, your responses will be treated as information only. It shall not be used as a proposal.

3. In accordance with FAR 15.209(c), the purpose of this RFI is to solicit technology experimentation candidates from research and development organizations, private industry, and academia for inclusion in future experimentation events coordinated by USSOCOM.

Contracting Office Address:
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MacDill AFB, Florida 33621-5323

Primary Point of Contact:
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